

Continuous functions – 7

Compactness, extrema and semicontinuity

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Theoretical minimum.

Def. f is *lower semicontinuous* at x_0 if $\liminf_{x \rightarrow x_0} f(x) \geq f(x_0)$, and upper semicontinuous if the reverse inequality holds; f is continuous exactly when it is both.

Theorem 1 (Weierstrass). A continuous function on a compact set is bounded and attains its extrema. More precisely, a lower semicontinuous function on a compact set attains its minimum, and an upper semicontinuous one attains its maximum.

Theorem 2 (coercivity). If f is continuous on $\mathbb{R}^n$ and $f(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, then f attains its minimum: the sublevel set $\{f \leq c\}$ is compact for a suitable c . This is how one proves existence of a minimum on a non-compact domain.

Fact 1. Dini's theorem: if continuous f_n increase pointwise to a continuous f on a compact set, the convergence is uniform. Compactness is essential in both hypotheses.

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1. Prove Weierstrass' theorem in the semicontinuous form, and give an example of a lower semicontinuous function on $[0, 1]$ which is unbounded above.
 2. Prove Dini's theorem and show by examples that it fails on $(0, 1)$ and fails for a discontinuous limit.
 3. (Iranian competition) Let (X, d) be a compact metric space and let $f: X \rightarrow X$ be surjective with $d(f(x), f(y)) \geq d(x, y)$ for all x, y . Prove that f is an isometry, and in particular that it is continuous.

4. (CIIM 2013) Let (X, d) be a connected compact metric space. Prove that there exists $\alpha \geq 0$ with the following property: for every integer $n > 0$ and all points $x_1, \dots, x_n \in X$ there is a point $x \in X$ with $\frac{1}{n} \sum_{i=1}^n d(x, x_i) = \alpha$.
5. Prove that a continuous function on $\mathbb{R}^n$ which tends to $+\infty$ at infinity attains its minimum, and use this to prove that every polynomial of even degree with positive leading coefficient attains a global minimum on $\mathbb{R}$.